A3 — Data and Query Design · Team 11 Transport & Parking

Course: 1305308 Platform Development · **Assignment 3:** Database Design (Group)
Team 11 — Campus Transport & Parking · Kanthi Phoosorn (6731503004)
Database: Cloudflare D1 `team11-transport-parking` (remote, region APAC)
API: https://team11-transport-parking.kanthiphs.workers.dev
Source: `team11-transport-parking-api/` — `migrations/`, `src/index.ts`, `scripts/api-demo.sh`
Evidence run: `20260915T083739Z` (UTC), full transcript in `evidence/api-demo-20260915T083739Z.txt`

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1. What the data has to support

The mini project (PRD Phase 1) answers one question on one screen: "*what's the next shuttle on my route, and where can I park near where I'm going?*" Four actors write or read data — student commuter, staff/permit holder, parking operator, shuttle dispatcher — and the PRD fixes three things the database must get right:

1. **One invariant.** `0 ≤ Availability.free ≤ ParkingZone.capacity` (PRD 3.2).
2. **One race.** Two operators PATCH the same zone at the same instant; neither update may be silently lost (PRD 3.2).
3. **One privacy rule.** A *Journey* is computed for the request and never stored, so no per-person movement history exists (PRD 3.2).

Everything in this document is shaped by those three lines.

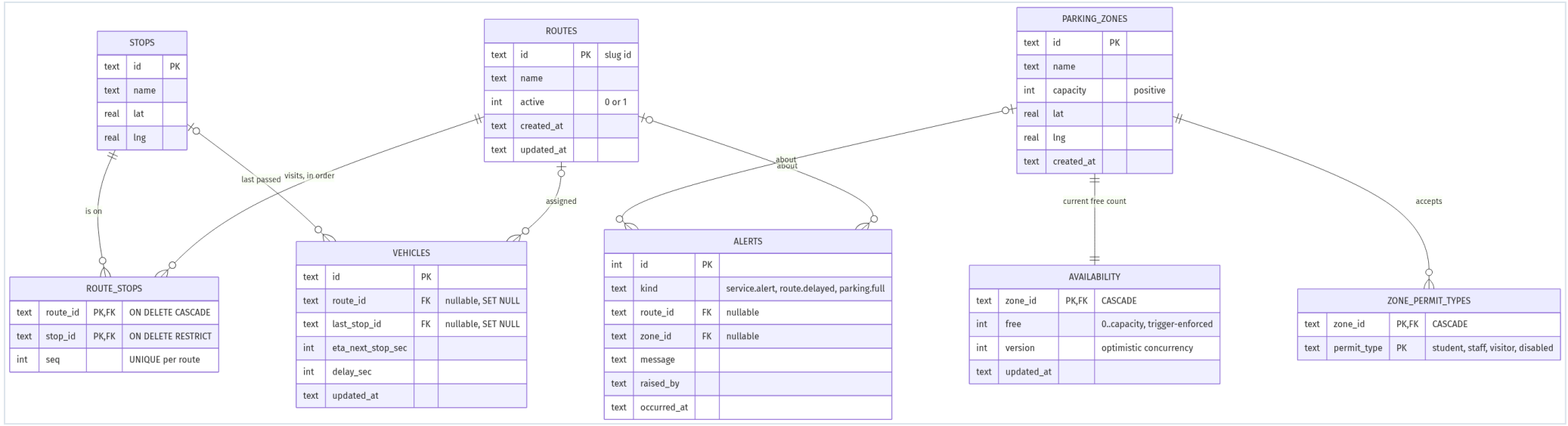
2. Why Cloudflare D1

The PRD's Phase 5 already picked "SQLite in dev, a relational store in prod" because the entities are relational and one SQL transaction solves the race. D1 *is* SQLite — served from Cloudflare's edge next to the Worker that runs our API (Assignments 2 and 5 already run there). So the choice was mostly about confirming that the other two options didn't buy anything the PRD asks for.

Need (from the PRD)	Cloudflare D1	Supabase (Postgres)	Firebase Firestore
Relational entities with real foreign keys (route ↔ stop, zone ↔ availability)	Yes — SQLite FKs, `CASCADE` / `RESTRICT`	Yes	No joins; references are manual, integrity is app code
Enforce `free ≤ capacity` *in the database*	Trigger (shown in §4)	CHECK / trigger	Only via security rules or app code
Solve the concurrent-PATCH race	One `UPDATE … WHERE version = ?`; D1 batch = transaction	Same, plus row locks	Transactions exist but on documents, not on a relational condition
Same platform as the API (no cross-service hop, no secrets to manage for a DB URL)	Yes — a binding on the Worker	No — separate service, connection pooling, latency	No — separate SDK and auth
Zero budget for a one-term project	Free tier: 5 GB, 5 M reads/day	Free tier, but projects pause after inactivity	Free tier
Migrations as versioned files in git	`wrangler d1 migrations` (used in §5)	Supabase CLI migrations	No schema, so no migrations
Fits the PRD's "SQLite → relational" plan	Exactly	Yes (different engine)	No

What we give up, honestly: D1 is single-writer per database and tops out at 10 GB; there is no Postgres-only feature set (no `LISTEN/NOTIFY`, no PostGIS). For four parking zones and three shuttle routes on one campus, none of that is a constraint we would hit in the life of this project. If it ever became one, the schema below is plain SQL and moves to Postgres without redesign.

3. ER diagram



Full-size image: `evidence/erd.png` · source: `evidence/erd.mmd`

Source: `evidence/erd.mmd` (Mermaid `erDiagram`; renders on GitHub).

Reading it: `ROUTE_STOPS` is the ordered many-to-many that replaces the PRD's `Route.stopIds[]` / `Stop.routeIds[]`; `ZONE_PERMIT_TYPES` replaces `ParkingZone.permitTypes[]`; `AVAILABILITY` is split from `PARKING_ZONES` because it has a different owner (the operator) and a different write rate; `ALERTS` records the events other teams consume. There is deliberately **no `JOURNEYS` table**.

4. Schema and the decisions behind it

Full DDL: `migrations/0001_schema.sql` (8 tables, 2 triggers, 2 indexes). The parts that carry the design:

Decision	Where	Why
IDs are short slugs (`R1`, `S-LIB`, `Z-GATE`) with a `CHECK (id GLOB '[A-Za-z0-9_-]*' AND length BETWEEN 1 AND 40)`	every entity table	Matches the PRD 4.3 contract regex, reads well in URLs and in events other teams receive
Array fields become join tables	`route_stops(route_id, stop_id, seq)`, `zone_permit_types`	Third normal form; "which routes serve this stop?" is one query, not a scan through JSON
`route_stops.seq` with `UNIQUE (route_id, seq)`	join table	A route's stop order is data, not client convention
`availability` is its own table, 1:1 with `parking_zones`	parking side	Ownership boundary from PRD 3.1 — only the operator writes it
`availability.version INTEGER`	parking side	Optimistic concurrency token; the `UPDATE` only applies if the caller's version is still current
Trigger `trg_availability_free_le_capacity_*` raising `'free exceeds zone capacity'`	parking side	SQLite `CHECK` can't read another table; the trigger makes the database refuse the write even if the API is bypassed
`ON DELETE CASCADE` on `route_stops.route_id`, `availability`, `zone_permit_types`; `ON DELETE RESTRICT` on `route_stops.stop_id`; `SET NULL` on `vehicles.route_id`	FKs	A deleted route takes its own stop list with it; a stop that routes still visit cannot be deleted; a vehicle whose route disappears becomes unassigned, not orphaned
`permit_type` is a constrained string, not a table	`zone_permit_types`	Permits belong to the Identity team (PRD 3.1); we reference the type, we never store permit holders
No `journeys` table	—	PRD 3.2 privacy by design; `GET /v1/journeys` returns `"stored": false`

The trigger, verbatim:

```
CREATE TRIGGER trg_availability_free_le_capacity_upd
BEFORE UPDATE OF free ON availability
BEGIN
  SELECT RAISE(ABORT, 'free exceeds zone capacity')
  WHERE NEW.free > (SELECT capacity FROM parking_zones WHERE id = NEW.zone_id);
END;
```

5. Migration and deployment to the remote database

Schema and seed are two versioned files applied with Wrangler's migration tool, so the remote database's state is reproducible from git.

```
migrations/0001_schema.sql   tables, FKs, triggers, indexes
migrations/0002_seed.sql     5 stops, 3 routes (1 inactive), 4 vehicles, 4 zones, availability

$ npx wrangler d1 migrations apply team11-transport-parking --remote
| 0001_schema.sql | ✅ |
| 0002_seed.sql   | ✅ |

$ npx wrangler d1 migrations list team11-transport-parking --remote
✅ No migrations to apply!
```

Proof from the remote database (run `20260915T083739Z`) — eight tables, the two triggers on `availability`, and no journeys table:

```
table_name        | triggers        routes | stops | route_stops | vehicles | zones | zone_permit_types |
availability | alerts
alerts            | 0               3      | 5     | 10          | 4        | 4     | 7                 | 4
| 0
availability      | 2
parking_zones     | 0
route_stops       | 0
routes            | 0
stops             | 0
vehicles          | 0
zone_permit_types | 0
```

The same migrations applied to a local D1 first; the trigger (`free exceeds zone capacity`), the `RESTRICT` (`FOREIGN KEY constraint failed` on deleting `S-LIB`) and the cascade (deleting `R9` left 0 `route_stops` rows) were all exercised at the SQL level before anything was deployed.

6. API

Deployed Worker: `https://team11-transport-parking.kanthiphs.workers.dev`. All routes are under `/v1`; error model `400 / 401 / 403 / 404 / 409 / 422` as in PRD 4.3. Operator and dispatcher authenticate with a bearer token (`OPERATOR_TOKEN`, `DISPATCHER_TOKEN`, stored as Worker secrets).

PRD	Method & path	Who	Notes
API-1	`GET /v1/routes` (`?active=all`)	public	with `stop_count`
API-2	`GET /v1/routes/:id`, `GET /v1/routes/:id/stops`	public	ordered by `seq`
—	`POST /v1/routes`, `PUT /v1/routes/:id`, `DELETE /v1/routes/:id`	dispatcher	route + ordered stops written in **one D1 batch** (transaction)
—	`GET /v1/stops`, `GET /v1/stops/:id`	public	with `route_ids`
—	`POST /v1/stops`, `PUT /v1/stops/:id`, `DELETE /v1/stops/:id`	dispatcher	delete → **409** if still on a route
API-3	`GET /v1/vehicles?routeId=`, `GET /v1/vehicles/:id`	public	
—	`POST /v1/vehicles`, `PATCH /v1/vehicles/:id`, `DELETE /v1/vehicles/:id`	dispatcher	`delaySec ≥ 300` raises `route.delayed`
API-4	`GET /v1/parking?permitType=`, `GET /v1/parking/:id`	public	zone + availability + permit types
API-5	`PATCH /v1/parking/:id` `{free, version}`	operator	**409** on stale version; `free → 0` raises `parking.full`
—	`POST /v1/parking`, `PUT /v1/parking/:id`, `DELETE /v1/parking/:id`	operator	create writes zone + availability + permits atomically
API-6	`POST /v1/alerts`, `GET /v1/alerts`	operator / dispatcher, public	
API-7	`GET /v1/journeys?origin=&dest=&permitType=`	public	computed, `"stored": false`

That is full create / read / update / delete on routes, stops, vehicles and parking zones, plus the two PRD-specific writes (availability, alerts).

7. Example requests and responses

All captured from the deployed API on `2026-09-15` (UTC). Bearer tokens are elided.

7.1 Read the shuttle network (API-1, API-2, API-3)

```
GET /v1/routes                                  → 200
{ "count": 2, "data": [
  { "id": "R1", "name": "Campus Loop",  "active": 1, "stop_count": 5, … },
  { "id": "R2", "name": "Dorm Express", "active": 1, "stop_count": 3, … } ] }

GET /v1/routes/R1/stops                         → 200
{ "route_id": "R1", "count": 5, "data": [
  { "id": "S-GATE", "name": "Main Gate",             "lat": 20.0424, "lng": 99.8951, "seq": 1 },
  { "id": "S-LIB",  "name": "Learning Resource Ctr", "lat": 20.0455, "lng": 99.893,  "seq": 2 },
  { "id": "S-E1",   "name": "E1 Building",           "seq": 3 }, { "id": "S-C5", "seq": 4 },
  { "id": "S-DORM", "name": "Lamduan Dormitories",   "lat": 20.0512, "lng": 99.8869, "seq": 5 } ] }

GET /v1/vehicles?routeId=R1                     → 200
{ "count": 2, "data": [
  { "id": "V-01", "route_id": "R1", "last_stop_id": "S-LIB", "eta_next_stop_sec": 180, "delay_sec": 0,  …
},
  { "id": "V-02", "route_id": "R1", "last_stop_id": "S-C5",  "eta_next_stop_sec": 240, "delay_sec": 60, … }
] }
```

7.2 Parking by permit (API-4)

```
GET /v1/parking?permitType=staff                  → 200
{ "count": 2, "data": [
  { "id": "Z-E1",  "name": "E1 Staff Parking",  "capacity": 40, "free": 12, "version": 1, "permit_types":
["disabled","staff"] },
  { "id": "Z-LIB", "name": "Library Car Park",  "capacity": 80, "free": 0,  "version": 1, "permit_types":
["staff","student"] } ] }
```

7.3 The happy path — a journey, computed and not stored (API-7)

```
GET /v1/journeys?origin=S-GATE&dest=S-DORM&permitType=student     → 200
{
  "origin":      { "id": "S-GATE", "name": "Main Gate" },
  "destination": { "id": "S-DORM", "name": "Lamduan Dormitories" },
  "shuttle": { "route_id": "R1", "route_name": "Campus Loop", "stops_to_ride": 4,
               "next_vehicle": { "id": "V-01", "last_stop_id": "S-LIB", "eta_next_stop_sec": 180,
"delay_sec": 0 } },
  "parking": { "id": "Z-DORM", "name": "Dormitory Car Park", "capacity": 200, "free": 133, "permit_types":
["student"] },
  "source": "deterministic",
  "stored": false,
  "computed_at": "2026-09-15T08:37:50.239Z"
}
```

The query behind `shuttle`: active routes that visit `origin` *before* `dest`, shortest ride first — a self-join on `route_stops` comparing `seq`. For `S-DORM → S-LIB` there is none (`R2` goes Dorm → E1 → Gate; `R9` is inactive), so the same call returns `"shuttle": null` with the nearest free zone still filled in.

7.4 The error model (PRD 4.3)

```
PATCH /v1/parking/Z-GATE  (no token)               → 401  { "error": "Bearer token required" }
PATCH /v1/parking/Z-GATE  (dispatcher token)       → 403  { "error": "Requires role: operator",
"your_role": "dispatcher" }
GET   /v1/parking/not%20valid                      → 400  { "error": "Invalid id" }
GET   /v1/parking/Z-NOPE                           → 404  { "error": "Zone not found" }
```

7.5 The race, for real (API-5 with optimistic concurrency)

Both operators read `version: 1`, then two `PATCH` requests were sent **concurrently** with that same version:

```
operator A  PATCH /v1/parking/Z-GATE  {"free":44,"version":1}
operator B  PATCH /v1/parking/Z-GATE  {"free":43,"version":1}

operator B → 200  { "id": "Z-GATE", "free": 43, "version": 2, "updated_at": "2026-09-15T08:37:52.502Z",
"events": [] }
operator A → 409  { "error": "Version conflict: another operator updated this zone first",
                    "your_version": 1, "current": { "free": 43, "version": 2 } }
```

One write won, the other was told exactly what it lost to. Nothing was silently overwritten — the PRD 3.2 race is closed by a single `UPDATE availability SET free = ?, version = version + 1 WHERE zone_id = ? AND version = ?`.

7.6 The invariant is the database's, not the API's

```
PATCH /v1/parking/Z-E1  {"free":999,"version":1}    → 422  { "error": "free must be <= capacity" }
```

The API only validates `free ≥ 0`; the 422 comes from the trigger's `RAISE(ABORT, 'free exceeds zone capacity')` being mapped to the PRD error model.

7.7 Edge-triggered events (PRD 4.4)

```
PATCH /v1/parking/Z-E1  {"free":0,"version":1}                      → 200  { …, "free": 0, "version": 2,
"events": ["parking.full"] }
PATCH /v1/vehicles/V-DEMO-…  {"lastStopId":"S-E1","delaySec":360}   → 200  { …, "delay_sec": 360, "events":
["route.delayed"] }

GET /v1/alerts                                                       → 200
{ "count": 3, "data": [
  { "id": 3, "kind": "service.alert", "route_id": "R1", "message": "Campus Loop diverted via C5 until
16:00", "raised_by": "dispatcher" },
  { "id": 2, "kind": "route.delayed", "route_id": "R2", "message": "Vehicle V-DEMO-… on R2 is 360s late",
"raised_by": "system" },
  { "id": 1, "kind": "parking.full",  "zone_id": "Z-E1", "message": "Zone Z-E1 is full",
"raised_by": "system" } ] }
```

`parking.full` fires only on the transition `free > 0 → 0` (setting an already-full zone to 0 again raises nothing). Delivering these to the Notification Hub as a signed webhook is Assignment 5.

7.8 Full CRUD on a parking zone (operator)

```
POST   /v1/parking  {"id":"Z-DEMO-…","name":"Demo Overflow
Lot","capacity":30,"lat":20.043,"lng":99.896,"permitTypes":["visitor","student"],"free":30}
       → 201  { "id": "Z-DEMO-…", "capacity": 30, "free": 30, "version": 1, "permit_types":
["student","visitor"], … }
PUT    /v1/parking/Z-DEMO-…  {"name":"Demo Overflow Lot (resized)","capacity":20,…,"permitTypes":
["visitor"]}
       → 200  { "capacity": 20, "free": 20, "version": 2, "permit_types": ["visitor"], … }      ← free
clamped to the new capacity
DELETE /v1/parking/Z-DEMO-…   → 200  { "deleted": "Z-DEMO-…", "note": "availability and zone_permit_types
cascaded" }
GET    /v1/parking/Z-DEMO-…   → 404
```

7.9 Full CRUD on routes and stops (dispatcher), and what `RESTRICT` protects

```
POST /v1/stops   {"id":"S-DEMO-…","name":"Demo Stop","lat":20.044,"lng":99.892}                    →
201
POST /v1/routes  {"id":"R-DEMO-…","name":"Demo Shuttle","stops":["S-GATE","S-DEMO-…","S-LIB"]}     →
201  (route + 3 route_stops rows, one batch)
PUT  /v1/routes/R-DEMO-…  {"name":"Demo Shuttle (rerouted)","active":true,"stops":["S-GATE","S-LIB"]}  →
200  (stops re-sequenced 1, 2)

DELETE /v1/stops/S-LIB       → 409  { "error": "Stop is still used by routes; remove it from them first",
"routes": ["R-DEMO-…", "R1"] }
DELETE /v1/stops/S-DEMO-…    → 200  (after it was taken off the route)
DELETE /v1/routes/R-DEMO-…   → 200  { "note": "route_stops cascaded; vehicles on this route now have
route_id = NULL" }
```

7.10 Vehicles (dispatcher)

```
POST   /v1/vehicles  {"id":"V-DEMO-…","routeId":"R2","lastStopId":"S-DORM","etaNextStopSec":120}   → 201
PATCH  /v1/vehicles/V-DEMO-…  {"lastStopId":"S-E1","etaNextStopSec":30,"delaySec":360}             → 200
(see 7.7)
DELETE /v1/vehicles/V-DEMO-…                                                                        → 200
```

8. Reproduce

```
cd team11-transport-parking-api
npm ci
npx wrangler d1 migrations apply team11-transport-parking --remote   # idempotent
npx wrangler deploy
bash scripts/api-demo.sh        # every call above, ~20 s, restores seed values at the end
```

Needs `wrangler login` and a `.dev.vars` with `OPERATOR_TOKEN` / `DISPATCHER_TOKEN` matching the Worker secrets.